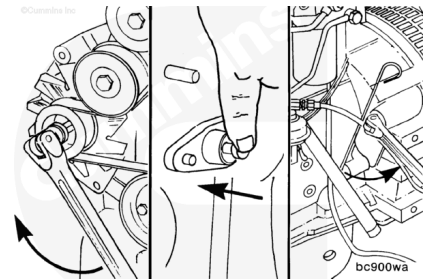


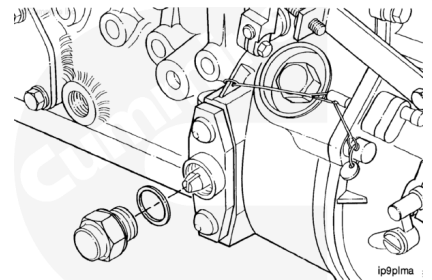
Trainings ▾

Install

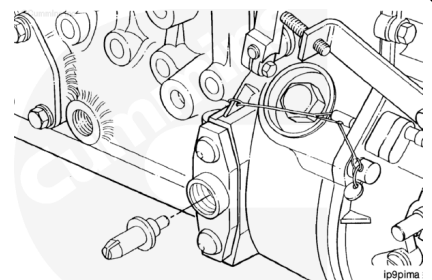
Make certain that the engine has cylinder Number 1 at top dead center.



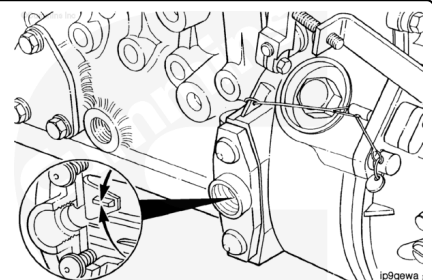
Remove the access plug.



Remove the timing pin.

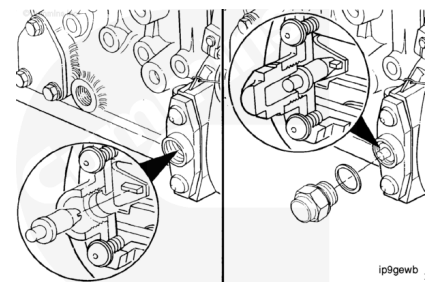


If the timing tooth is **not** aligned with the timing pin hole, rotate the fuel injection pump shaft until the timing tooth aligns.

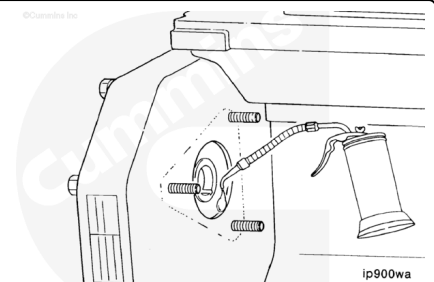


Reverse the position of the timing pin so the slot of the timing pin will fit over the timing tooth in the pump.

Install and secure the timing pin with the access plug.

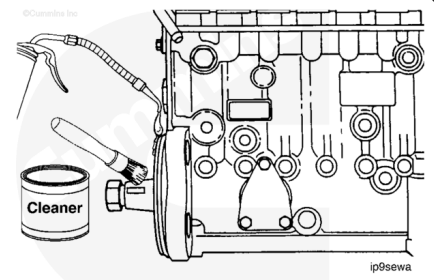


Use a 50/50 mixture of clean lubricating engine oil and STP, or equivalent, to lubricate the gear cover housing to make certain that the fuel injection pump will slide into the gear cover housing easily.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

⚠ CAUTION ⚠

The fuel injection pump drive gear inside diameter and the shaft outside diameter must be clean and dry before installing the gear. Failure to do so can result in slipped timing.

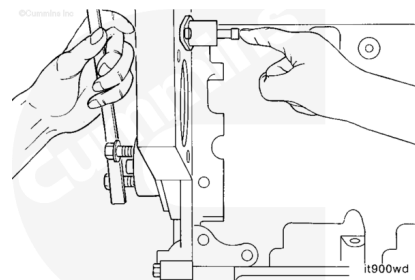
Note : Before installing the fuel pump drive gear, clean the injection pump shaft and gear tapers with residue-free cleaner, Part Number 3824510, by spraying into the gap between the shaft and the gear. Dry with compressed air.

Note : The in-line fuel injection pump driveshaft has a provision for a Woodruff key: however, it is not required. Timing mark alignment is not required for the in-line drive gear.

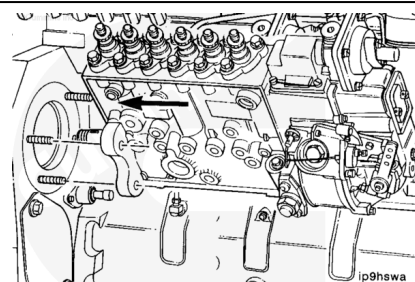
Note : Make certain that the engine has cylinder Number 1 at top dead center.

Make certain that the o-ring seals for the fill orifice and pilot are correctly installed and are **not** damaged.

Install new pilot o-ring.

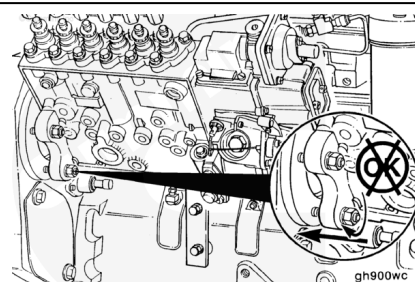


Push the pump forward until the mounting flange and o-ring are properly fitted into the gear housing bore.



⚠ CAUTION ⚠

Do not attempt to pull the pump flange into the gear housing with the mounting nuts as damage to housing can occur.

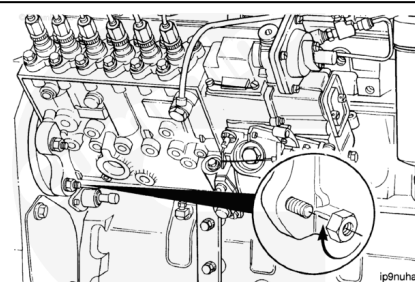


Install the mounting nuts.

Torque Value: 43 n•m [32 ft-lb]

Install the support bracket (if equipped).

Torque Value: 32 n•m [24 ft-lb]

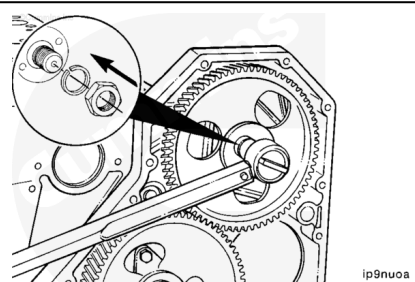


Install the fuel injection pump drive gear onto the fuel pump shaft.

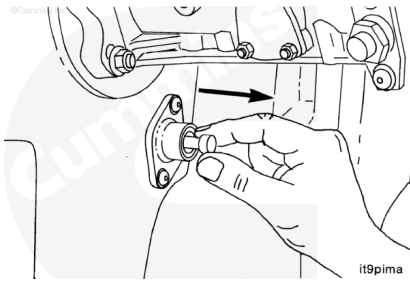
Install the retaining nut and washer.

Torque Value: 10 to 15 n•m [89 to 133 in-lb]

To prevent damage to the timing pins, do **not** exceed the torque value given. This is **not** the final torque value for the retaining nut.



Disengage the engine timing pin.



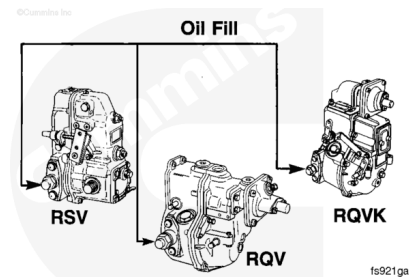
⚠ CAUTION ⚠

The governor housing must be prelubricated before engine operation. Failure to do so can result in premature governor wear.

Remove the access plug.

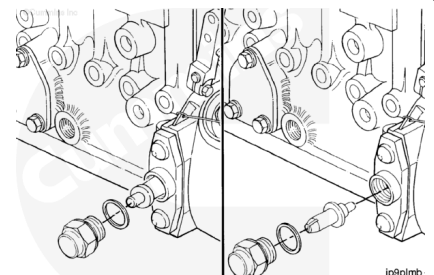
Add the following quantity of clean lubricating engine oil:

- RSV 450 mL [0.48 qt]
- RQV 750 mL [0.79 qt]
- RQVK 750 mL [0.79 qt]



Remove the fuel injection pump timing pin plug, reverse the position of the timing pin, and install the timing pin, plug, and sealing washer.

Torque Value: 27 N•m [20 ft-lb]



Tighten the fuel injection pump drive nut.

Torque Value:

"A" Pump

1.85 N•m [63 ft-lb]

Torque Value:

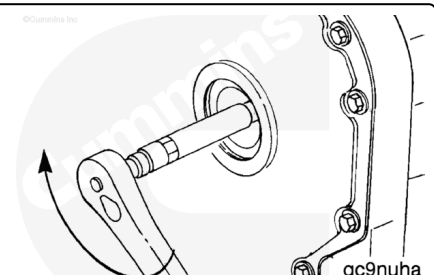
P3000 and P7100

1.195 N•m [144 ft-lb]

Torque Value:

Nippondenso

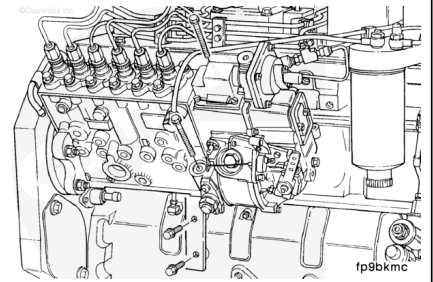
1.123 N•m [91 ft-lb]



Install the fuel injection pump mounting bracket capscrews.

Tighten all capscrews by hand for proper alignment.

Torque Value: 24 n•m [18 ft-lb]



Connect the external oil feed line at the inboard side of the fuel injection pump (if applicable) and the main oil rifle.

Connect the external oil feed line at the rear of the pump or AFC latchout if applicable.

